Pre-Analysis Plan: Effects of Having Daughters on Gender Attitudes and Behavior

Team Code: RT

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Overview

This Pre-Analysis Plan (PAP) outlines the empirical strategy for testing four causal hypotheses using 5% of the SOEP(The Socio-Economic Panel) dataset which we will have access to after submitting this PAP to the project coordinators. Each hypothesis explores how having daughters affects parental attitudes and behaviors regarding gender and politics. The analysis will employ fixed-effects panel regression models and present one regression table per hypothesis.

Hypotheses and Model Specifications

We estimate individual fixed-effects models to test how changes in child gender composition—particularly the presence of daughters—affect gender-related attitudes, political views, and household behavior over time. All models rely on within-individual variation and control for time-invariant characteristics by construction.

General Specification

The core fixed-effects specification is as follows:

$$Y_{it} - \bar{Y}i = \beta(PCGit - P\bar{C}Gi) + \gamma(Xit - \bar{X}i) + \delta_1(PCGit \times Country_i) + \delta_2(PCG_{it} \times Year_t) + \epsilon_{it} \quad (1)$$

Where:

- Y_{it} is the outcome of interest for individual i at time t: gender equality attitudes (H1, H4), political preferences (H2), or household gender behavior (H3),

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- PCG_{it} is a set of variables representing the four different parent-child gender combinations, for more information referring to Independent Variables section,
- X_{it} is a vector of time-varying control variables (e.g., age, squared age, marital status, highest education qualification, income, religiosity, year fixed effects, country fixed effects),
- $PCG_{it} \times Country_i$ is an interaction between parent-child gender combinations and country dummies,
- $PCG_{it} \times Year_t$ is an interaction between parent-child gender combinations and year dummies,
- $\bar{Y}_i, \bar{PCG}_i, \bar{X}_i$ are person-specific means of the respective variables,
- β, γ, δ_1, and δ_2 are coefficients of interest,
- ϵ_{it} is the error term clustered at the individual level.

Sample Selection

We plan to apply the following criteria to construct the analytical sample:

- Restrict the sample to person-year observations in which respondents were aged between 18 and 50 years (inclusive), to focus on prime childbearing and child-rearing years.
- Exclude person-year observations with missing data on any model variable. The majority of missingness stems from non-response to survey items used to construct the dependent variables.
- Exclude respondents who were observed only once in the panel, as fixed-effects models require at least two observations per individual.
- Retain respondents who were already parents at the beginning of the panel or who entered as parents after baseline. Likewise, childless individuals who remain childless throughout the panel are retained, although they do not contribute to the estimation of within-person parenthood effects.
- Not impute missing outcome values.

The final estimation sample consists of unbalanced panel data with at least two observations per individual.

Dependent Variable

We use different dependent variables for each hypothesis, aligned with the specific attitudinal or behavioral domain being tested:

- **H1 and H4 – Gender Equality Attitudes (Y_{it})**: Measured using a composite index based on responses to survey questions capturing support for gender equality . Items are recoded where necessary so that higher values reflect more egalitarian attitudes. The index is standardized within waves of surveys to ensure comparability across time.

- **H2 – Political Preferences (Y_{it})**: Based on respondents' self-placement on a left–right ideological scale ranging from 0 (far left) to 10 (far right). The scale is reverse-coded so that higher values indicate more left-leaning political preferences, consistent with the literature linking egalitarian values to progressive ideology.

- **H3 – Household Gender Equality in Behavior (Y_{it})**: The share of total household labor (including chores and care work) performed by the male partner in couple households. Constructed as:

$$Y_{it} = \frac{\text{Male housework hours}_{it}}{\text{Male housework hours}_{it} + \text{Female housework hours}_{it}}$$

  where housework hours include routine domestic tasks such as cooking, cleaning, laundry, and caring for children or other household members. The measure ranges from 0 (completely unequal) to 1 (fully equal), with higher values indicating greater male contribution and more balanced division of labor.

All dependent variables are time-varying and measured at the individual level. Where applicable, responses are harmonized across waves to allow longitudinal analysis using fixed-effects models.

Independent Variables

To construct the key explanatory variables, we begin by creating a time-varying dummy indicator for parenthood. This "base" parenthood variable takes the value 1 if the respondent has been observed to have a firstborn child during the life of the panel, and 0 otherwise.

To distinguish parenthood by gender of the parent, we split this base variable into two binary indicators.

- **Father**: equals 1 if the respondent is male and has had his firstborn child during the observation window; 0 otherwise.

- **Mother**: equals 1 if the respondent is female and has had her firstborn child during the observation window; 0 otherwise.

We then further differentiate parenthood by the gender of the firstborn child. Specifically, we construct two additional dummy variables:

- **Daughter**: equals 1 if the respondent has had a first-born girl; 0 otherwise.

- **Son**: equals 1 if the respondent has had a firstborn boy; 0 otherwise.

Building on these classifications, we define four mutually exclusive and exhaustive indicators combining parent gender and child gender.

- **Male Parent of Daughter**: equals 1 if the respondent is male, has had his firstborn child, and the child is a girl.

- **Male Parent of Son**: equals 1 if the respondent is male, has had his firstborn child, and the child is a boy.
- **Female Parent of Daughter**: equals 1 if the respondent is female, has had her firstborn child, and the child is a girl.
- **Female Parent of Son**: equals 1 if the respondent is female, has had her firstborn child, and the child is a boy.

Each of these indicators remains equal to zero in which the individual does not meet the criteria and becomes one from the year in which the first qualifying child is observed onward.

Because the fixed-effects estimation relies on within-individual variation over time, only respondents whose parenthood status changes during the panel (i.e., those who transition into parenthood) contribute to the estimation of coefficients on these variables.

Hypothesis 1: Having daughters increases support for gender equality (attitudes)

To test whether having a daughter increases support for gender equality, we compare respondents who become parents of daughters to those who become parents of sons, separately for men and women. The model includes the following four mutually exclusive dummy variables:

- Male Parent of Daughter
- Male Parent of Son
- Female Parent of Daughter
- Female Parent of Son (reference category)

Key parameters: We test whether parents of daughters—both male and female—have higher gender equality attitudes than parents of sons. Specifically:

$$\beta_{\text{MPD}} > \beta_{\text{MPS}} \quad \text{and} \quad \beta_{\text{FPD}} > 0$$

where β_{MPD}, β_{MPS}, and β_{FPD} refer to the coefficients on Male Parent of Daughter, Male Parent of Son, and Female Parent of Daughter, respectively.

Hypothesis 2: Having daughters is associated with more left-leaning political preferences

We use the same set of parent-child gender indicators in a fixed-effects model where the dependent variable is self-reported left–right political orientation (reverse-coded).

Key parameters: As in H1, we compare:

$$\beta_{\text{MPD}} > \beta_{\text{MPS}} \quad \text{and} \quad \beta_{\text{FPD}} > 0$$

to assess whether parents of daughters—especially men—become more politically progressive than parents of sons.

Hypothesis 3: Having daughters leads to greater gender equality in household behavior

For this hypothesis, the dependent variable is the share of household labor performed by the male partner. The model again includes the four parent-child gender indicators.

Key parameter: We test whether men who become fathers of daughters show greater involvement in housework than those who have sons:

$$\beta_{\text{MPD}} > \beta_{\text{MPS}}$$

Hypothesis 4: The increase in gender equality attitudes due to having daughters is greater for men than for women

This hypothesis directly tests whether male respondents react more strongly to having daughters than female respondents. Operationally, this is captured by testing whether the difference between daughters and sons is larger for men than for women.

Key comparison:

$$(\beta_{\text{MPD}} - \beta_{\text{MPS}}) > (\beta_{\text{FPD}} - 0)$$

That is, we test whether the attitudinal change from having a daughter (vs. a son) is larger among male than female respondents.

Estimation and Inference

- **Estimation method**: We use linear fixed-effects regression (OLS) for continuous outcomes. For outcomes that are ordinal or bounded (e.g., household labor share), we will test robustness using ordered logit/probit or fractional logit models.
- **Standard errors**: All models will use robust standard errors. When appropriate (i.e., with repeated observations), errors will be clustered at the individual level to account for within-person autocorrelation.
- **Inference and reporting**: Each hypothesis will be tested via regression models specified above. Coefficients, robust standard errors, t-statistics, and p-values will be presented in a single table per hypothesis. Marginal effects from interaction terms (e.g., H4) will be computed and visualized where necessary.

- **Sensitivity checks**: We will conduct robustness checks including alternative codings of child gender (e.g., number of daughters), and exclusion of outliers in the dependent variables.

Conclusion

This pre-analysis plan specifies the hypotheses, variables, sample restrictions, modeling strategies, and inferential procedures for testing the impact of having daughters on attitudes, political preferences, and household behavior. By predefining all analytic steps, we aim to ensure transparency, prevent post hoc theorizing, and limit researcher degrees of freedom. The analysis will follow the outlined specifications exactly unless deviations are formally documented and justified.